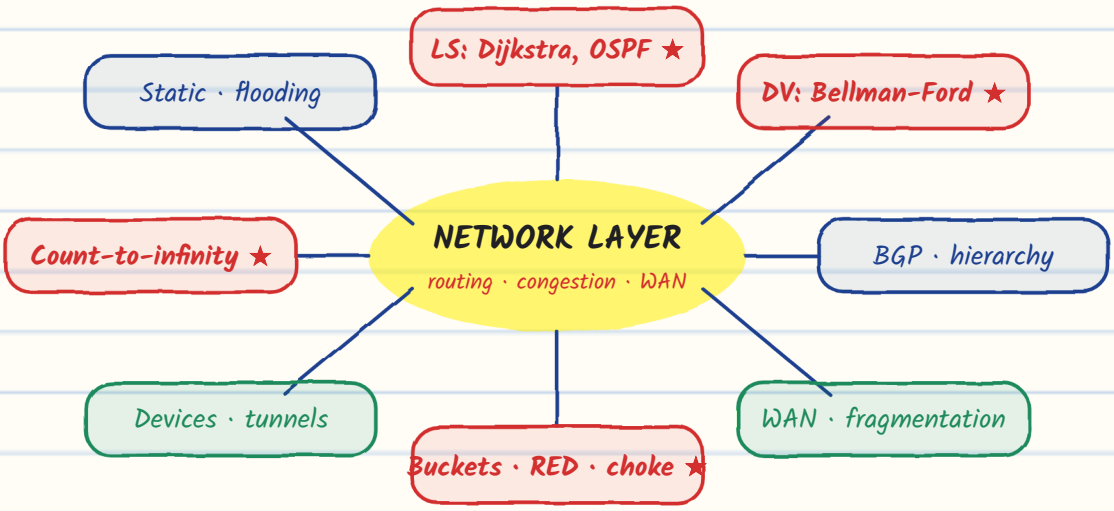
Date: 09.10.2026
topper's notes

Routing, Congestion Control & WAN

Dijkstra · Bellman-Ford · BGP · buckets · devices · X.25/FR/ATM/MPLS · fragmentation: exam-ready in 12 sheets



↑ the whole lesson on one page: red = high-yield, green = applied

➜ Syllabus coverage

"Computer Networking and Security: … Routing algorithm; Congestion control; Internetworking, WAN; Fragmentation; …". Neighbouring notes: layers & LANs (IEEE 802), IPv4/IPv6 header and addressing, ARP (network layer notes), TCP's congestion window (transport notes).

High-yield list

① DV → Bellman-Ford
② LS → Dijkstra
③ RIP: 16 hops = ∞
④ split horizon
⑤ $S = C/(M - \rho)$
⑥ offset ÷ 8 bytes
⑦ ATM cell 53 B

Legend yellow = key term · red circle = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

➜ Contents

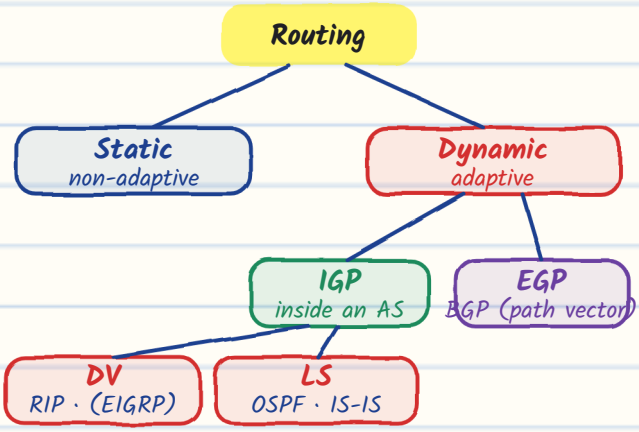
✓ Routing basics, flooding ..... p. 2
✓ Link state, Dijkstra, OSPF, hierarchy … p. 3
✓ Distance vector, RIP, count-to-∞ … p. 4
✓ BGP, broadcast, congestion ..... p. 5
✓ Choke packets, RED, shaping ..... p. 6
✓ Leaky & token bucket, devices … p. 7
✓ Tunnelling, switching ..... p. 8
✓ WAN: leased lines, X.25, FR, ATM, MPLS … p. 9
✓ IPv4 fragmentation ..... p. 10
✓ Night-before sheet ..... p. 11
✓ Practice MCQs + written ..... p. 12

1 Routing: the Big Picture

MCQ + Written

Definition A routing algorithm is the network-layer (layer 3) software that decides on which output line an incoming packet should be sent. **Routing** = building the routing table (control plane); **forwarding** = looking up each packet in that table and sending it out (data plane).

Optimality principle If J is on the optimal path from I to K, the optimal path from J to K follows the same route. So all optimal routes to one destination form a tree: the sink tree (no loops).



the family tree of routing

	Static (non-adaptive)	Dynamic (adaptive)
Routes set by	administrator, by hand	protocol exchanges between routers
On failure · overhead · use	no reaction · none · small/stub networks	re-routes itself · bandwidth, CPU · large networks

Metrics: hop count (RIP) · bandwidth · delay · load · reliability · MTU · cost (OSPF) · AS path / policy (BGP). **Lower metric = better** route.

Memory trick "**RIP counts hops, OSPF counts cost, BGP counts ASes** (and obeys policy)".

Exam trap **Metric** compares routes **within one protocol**; **administrative distance** (AD, trustworthiness) chooses **between protocols**, lower wins. Cisco AD: connected 0 · static 1 · eBGP 20 · EIGRP 90 · OSPF 110 · IS-IS 115 · RIP 120 · iBGP 200. So an OSPF route beats a RIP route whatever the hop count.

2 Flooding

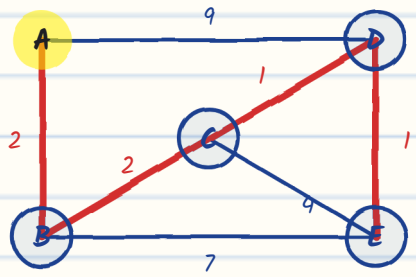
A static algorithm: send every packet out on **every line except the one it arrived on**. Robust, and always finds the shortest path (every path is tried), but floods the network with duplicates. Damp it with a **hop counter** (start at the diameter), **sequence numbers** (drop copies already seen) or **selective flooding**. Used to distribute link-state packets.

3 Link State Routing & Dijkstra high-yield

Idea Every router learns the complete topology map and runs Dijkstra's algorithm (SPF) on it itself: "routing by map".

1. **Discover neighbours** (HELLO) · **measure the cost** to each · **build a link state packet** (LSP/LSA: ID, sequence no., age, neighbours + costs)
2. **Flood** the LSP to **all** routers · **run Dijkstra** on the full map → shortest-path tree → routing table

→ Worked example: shortest paths from A



red = final shortest-path tree

Step	Permanent set	B	C	D	E
0	{A}	2,A	∞	9,A	∞
1	{A,B}	✓	4,B	9,A	9,B
2	{A,B,C}	✓	✓	5,C	8,C
3	{A,B,C,D}	✓	✓	✓	6,D

"4,B" = distance 4 via previous node B. Each step makes the **smallest tentative** node permanent and relaxes its neighbours. Every route from A leaves via next hop **B**; A→E = A-B-C-D-E, cost 6 (not the direct-looking A-D-E = 10).

OSPF facts

RFC 2328 (v2), v3 for IPv6
runs **directly on IP**: protocol **89**
cost = 10^8 / bandwidth (100 Mbps → 1)
areas; backbone = **area 0**; ABR, ASBR, DR/BDR
224.0.0.5 all routers · 224.0.0.6 DRs
Hello 10 s, dead 40 s · AD 110

Memory trick OSPF in numbers: "**89** · **110** · **area 0** · **5/6** · **10/40**" = protocol, AD, backbone, multicast ends, hello/dead.

IS-IS: OSI link-state twin, runs over layer 2, not IP. Dijkstra: $O(N^2)$.

Exam trap Dijkstra fails with **negative** edge costs; Bellman-Ford handles them (but not negative cycles).

4 Hierarchical Routing

Routers are grouped into **regions** (areas, ASes); a router keeps every router of its own region but only **one entry per other region**. Smaller tables, slightly longer paths. Optimum: **ln N** levels, **e·ln N** entries.

Worked example ✍ 720 routers. Flat: **720** entries. 24 regions × 30 routers: 30 + 23 = **53**. 8 clusters × 9 regions × 10 routers: 10 + 8 + 7 = **25**.

5 Distance Vector Routing & Bellman-Ford

high-yield

Idea Each router keeps the best known **distance** (and line) to every destination and swaps this vector **only with its neighbours**, periodically. "Routing by rumour"; ARPANET until 1979, then RIP.

Bellman-Ford equation $D_x(y) = \min_v \{ c(x, v) + D_v(y) \}$ over all neighbours v of x

Worked example ✍️ Router X: links X–A = 2, X–B = 5, X–C = 1. Received vectors:

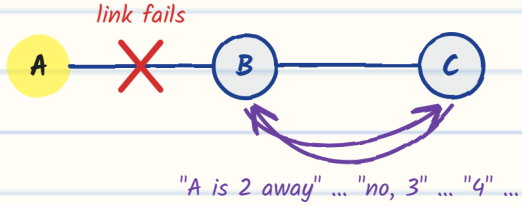
from	to Y	to Z
A	7	3
B	2	9
C	8	6

Y: min(2+7, **5+2**, 1+8) = **7 via B**
Z: min(**2+3**, 5+9, 1+6) = **5 via A**

RIP facts

metric = hop count; max **15**, **16 = ∞**
full table every **30 s** · UDP port **520**
invalid 180 s · hold-down 180 s · flush 240 s
v1 classful, broadcast · v2 classless, multicast **224.0.0.9**, auth
RIPng (IPv6): UDP 521 · AD 120

6 Count-to-Infinity & Its Fixes



exchange	B's dist to A	C's dist to A
before failure	1	2
1	3 (via C!)	2
2 · 3 · 4	3 · 5 · 5	4 · 4 · 6
finally	**16 = ∞**	**16 = ∞**

C still advertises "A at 2" (a route through B itself) and B believes it. Good news travels fast, bad news travels slowly.

Fix	How it works
Split horizon · poison reverse	never advertise a route back on the interface it came from · or send it back with metric 16 (∞)
Route poisoning · triggered updates	a failed route is sent at once with metric 16, not after 30 s
Hold-down timer	after a route goes down, ignore worse news about it for 180 s

Exam trap Split horizon cures the two-router loop but **not** loops of three or more routers; RIP's small ∞ (16) only caps the count.

	Distance vector	Link state
Knows · algorithm	neighbours' distances · Bellman-Ford	whole topology · Dijkstra
Sends	whole table, to neighbours, periodically	own links (LSA), to all routers, on change
Convergence · CPU	slow, count-to-∞ · low, small nets	fast, loop-free · high, large nets
Examples	RIP, IGRP; EIGRP = "advanced DV / hybrid" (DUAL)	OSPF, IS-IS

7 BGP: Routing Between Autonomous Systems MCQ

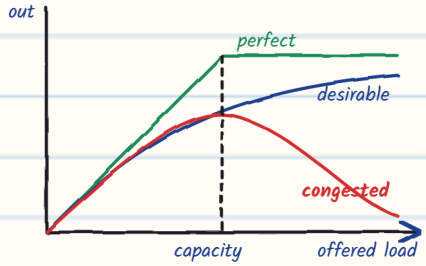
- Path vector protocol; BGP-4 is the Internet's inter-domain (EGP) protocol
- runs over **TCP port** 179; eBGP between ASes, iBGP inside one
- a route carries its full **AS_PATH**; a router rejects a path that holds its own AS → **no loops**
- chooses by **policy** (LOCAL_PREF, AS_PATH length, MED), not by cost · messages: OPEN, UPDATE, KEEPALIVE, NOTIFICATION

8 Broadcast & Multicast Routing

Broadcast: separate copies (wasteful) · flooding · multidestination · spanning tree (optimal) · reverse path forwarding (RPF): forward a copy only if it arrived on the line this router uses to reach the source. **Multicast** (224.0.0.0/4, IGMP): source-based trees (DVMRP, PIM-DM) or shared core-based trees (CBT, PIM-SM).

9 Congestion & Its Control high-yield

Definition Congestion: too many packets in (part of) the network, so delay rises and throughput falls. **Congestion control** is global (the whole network); **flow control** is point-to-point (fast sender, slow receiver).



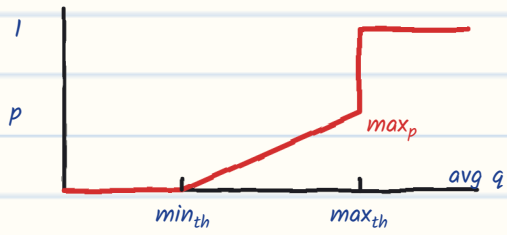
packets delivered vs offered

Causes: bursty input converging on one line, slow routers, too few buffers, low-bandwidth lines.
Infinite buffers make it **worse**: packets queue so long they time out and are retransmitted, adding load. Past capacity output can fall towards 0: congestion collapse.

	Open loop (prevention)	Closed loop (feedback, reaction)
Idea	good design so congestion never starts	monitor, signal, then adjust
Tools	provisioning, **admission control**, traffic shaping (buckets), retransmission/ACK/discard policies	**choke packets**, back-pressure, ECN, RED, TCP's AIMD window (transport notes)

- **Admission control**: in virtual-circuit networks, refuse a new VC that would congest the network (the telephone "busy" tone).
- **Choke packet**: a congested router sends a packet **back to the source**, which cuts its rate (e.g. by half). **Hop-by-hop back-pressure**: the choke takes effect at every router it passes, so relief is immediate. (ICMP Source Quench is now deprecated.)
- **ECN**: the router **marks** 2 bits in the IP header instead of dropping; the receiver echoes it (TCP ECE flag), the sender slows down.
- **Load shedding**: drop packets. Wine (old is better: file transfer, drop the new) vs milk (new is better: real-time media, drop the old).

RED (Random Early Detection, Floyd & Jacobson 1993): the router tracks the **average** queue length. Below min_{th}: no drop; between thresholds: drop with probability rising to max_p; above max_{th}: drop all. Dropping **early and at random** slows a few TCP senders and avoids tail-drop's global synchronisation.



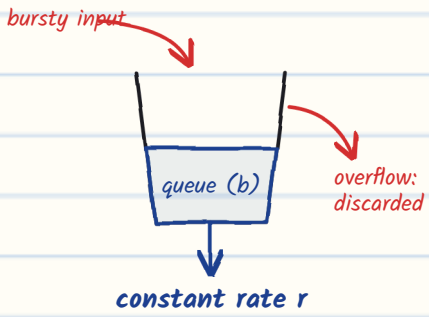
RED drop probability

Exam trap Congestion control ≠ flow control: a fast PC sending to a slow printer on an idle link is a flow-control problem; many hosts on one overloaded link is congestion.

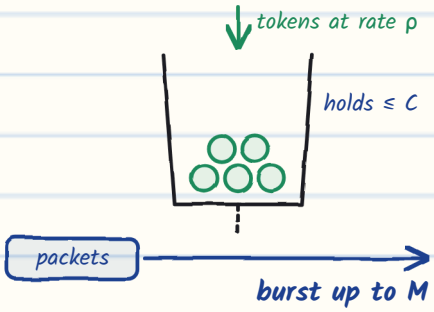
10 Traffic Shaping: Leaky & Token Bucket

written favourite

Traffic shaping regulating the **rate and burstiness** of the traffic a host sends, as agreed with the network (open loop). **Policing** = the network checking it and dropping or marking the excess.



leaky bucket: smooth output



token bucket: bursts allowed

	Leaky bucket	Token bucket
Output	**constant** rate r, however bursty the input	average rate ρ, bursts up to M allowed
Idle host · when full	gains nothing · discards **packets**	saves up to C tokens · discards **tokens**
Max data in time t	r·t	C + ρt (and ≤ M·t)

Burst length Full bucket C bytes, token rate ρ, maximum output rate M. During a burst of S seconds: C + ρS = MS ⇒ **S = C / (M − ρ)**

Example 1 ✍️ C = 250 KB, M = 25 MB/s, ρ = 2 MB/s: S = 250 KB / 23 MB/s ≈ **11 ms** at full speed, then 2 MB/s.

Example 2 ✍️ 1 MB burst (25 MB/s for 40 ms) into a leaky bucket, output 2 MB/s: output lasts **500 ms**; bucket must hold ≈ 0.92 MB.

Memory trick **Leaky** = a dripping tap: always the same drip. **Token** = a savings account: save while idle, spend in a burst, never more than the balance C.

11 Internetworking & Its Devices high-yield

Definition *Internetworking* = joining dissimilar networks (addressing, packet size, service, QoS) so they act as one; layer-3 routers running IP glue the Internet together.

Device	Layer	Works on	Collision domains	Broadcast domains
Repeater	1	bits: regenerates the signal	1 (extends it)	1
Hub	1	bits: multiport repeater	1 for all ports	1
Bridge	2	frames: MAC filtering	1 per port	1
Switch	2	frames: multiport bridge	**1 per port**	1 (per VLAN)
Router	3	packets: IP address	1 per port	**1 per port**
Gateway	4–7	protocol conversion	–	–

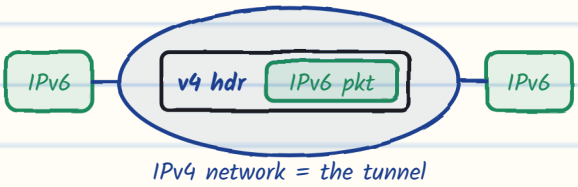
- **Bridge**: **backward learning** of MAC → port from source addresses; floods unknown destinations; STP (802.1D) removes loops.
- **Switch**: store-and-forward (checks CRC) or cut-through (lower latency); full duplex, no collisions; a layer-3 switch also routes.
- **Gateway**: converts between dissimilar protocols (e-mail, VoIP); the TCP/IP "default gateway" is just the router.

Worked example ✍ A router with 2 interfaces: one to a 24-port switch with 20 PCs, one to a hub with 6 PCs. Collision domains: 20 + 1 (switch–router link) + 1 (whole hub side) = 22. Broadcast domains: **2** (one per router interface).

Exam trap A **switch** breaks collision domains but **not** broadcast domains; only a **router** (or VLANs) splits broadcast domains. A hub is a layer-1 device even though it has many ports.

→ Tunnelling

To cross a network of a **different type**, the whole packet travels as the **payload** of the foreign network's packet between two multiprotocol routers. Examples: IPv6-in-IPv4 (protocol 41), GRE (protocol 47), IPsec VPN.



12 Circuit vs Packet Switching

MCQ + Written

	Circuit	Datagram	Virtual circuit
Idea	dedicated path set up first	each packet routed alone	path fixed at setup, links shared
Packet carries	–	full address	short VC number
Order · router fails	in order · call lost	may be out of order · only its packets lost	in order · all VCs through it end
QoS, congestion	easy (reserved)	difficult	easy if resources reserved
Examples	PSTN, ISDN	IP	X.25, Frame Relay, ATM, MPLS

Delay formulas message x bits, k hops, bit rate b, propagation d per hop, setup s, packet p bits (headers ignored): circuit $T = s + x/b + k\cdot d$ · packet $T = x/b + (k-1)\cdot p/b + k\cdot d$. Packet switching wins when $s > (k-1)\cdot p/b$.

Example ✍ $x = 10^6$ bits, $b = 1$ Mbps, $k = 4$, $d = 5$ ms, $s = 200$ ms, $p = 10\,000$ bits: circuit $0.2 + 1 + 0.02 =$ **1.22 s**; packet $1 + 3 \times 0.01 + 0.02 =$ **1.05 s**.

Exam trap A virtual circuit is **packet** switching (links are shared), not circuit switching; it only fixes the route at setup.

13 WAN Technologies high-yield

A **WAN** spans cities, countries, continents, usually over a carrier's lines: leased lines, circuit-switched ISDN, packet-switched X.25 / Frame Relay / ATM, MPLS, Internet VPN (IPsec), satellite/cellular backup.

Leased line	Rate	Channels
T1 (USA)	1.544 Mbps	24 × 64 k
E1 (Europe, BD)	2.048 Mbps	32 (30 usable)
T3 / E3	44.736 / 34.368	672 / 480
OC-3 = STM-1	155.52 Mbps	SONET/SDH
OC-12 = STM-4	622.08 Mbps	

ISDN: BRI = 2B + D = 2 × 64 + 16 = **144 kbps**; PRI = 23B + D (T1) or 30B + D (E1). T1 = 24 × 64 kbps + 8 kbps framing = 1.544 Mbps. OC-1 = 51.84 Mbps; OC-n = n × 51.84.

	X.25	Frame Relay	ATM
Era · body	1976, CCITT/ITU-T	early 1990s	1990s, B-ISDN core
Layers	3 (X.21, LAPB, PLP)	2 only (LAPF)	physical, ATM, AAL
Unit	packet (variable)	frame (variable)	**53-byte cell**
Error control	**hop-by-hop** (noisy lines)	none: bad frames dropped	HEC on header only
VC identifier	LCN	**DLCI** (10 bits)	**VPI/VCI**
Speed	≤ 64 kbps (later 2 Mbps)	56 kbps – 45 Mbps	155.52 / 622.08 Mbps
Extra	PAD for terminals	CIR, DE bit, FECN/BECN	CBR, VBR, ABR, UBR



header 53 bytes in all (≈ 9.4 % overhead)

ATM cell



32-bit MPLS shim header (bits)

MPLS label entry

ATM: 48 bytes was a compromise (USA wanted 64, Europe 32). UNI header: GFC 4, VPI 8, VCI 16, PT 3, CLP 1, HEC 8 bits. The ATM network layer and AAL are in their own notes.

MPLS ("layer 2.5", RFC 3031): the ingress **LER** classifies a packet into a FEC and **pushes** a label; core **LSRs swap** labels (exact match, fast); the egress **pops** it. The path is an LSP. Carriers use it for traffic engineering, QoS and L3 VPNs; it carries any protocol.

Memory trick **X.25 checks every hop, Frame Relay checks none, ATM chops everything into 53.** Frame Relay = "X.25 with the error control thrown out, because fibre is clean".

Exam trap ATM's HEC protects only the **5-byte header**, not the payload. Frame Relay does **detect** errors (FCS) but never **corrects** them: it discards the frame.

14 IPv4 Fragmentation

written favourite

Why A datagram bigger than the next link's **MTU** (largest frame payload) is split into **fragments**, each a full IP datagram with its own header. IPv4: the source **or any router** fragments; **reassembly only at the final destination**.

Header field (bits)	Role
Identification (16)	same in all fragments
DF (1) · MF (1)	don't fragment · more follow
Fragment offset (13)	in **8-byte** units
Total length (16)	header + data ≤ 65 535

Link	MTU (B)
Ethernet · PPP	1500
802.11 · FDDI	2304 · 4352
X.25 (typical)	576
IPv6 minimum	1280

Worked example 1 ✍ Total length 4000 B (20-byte header), next link MTU 1500.

1. Data = 4000 − 20 = 3980 B. Max data per fragment = 1500 − 20 = 1480 (a multiple of 8 ✓)
2. Split 3980 = 1480 + 1480 + 1020

Frag	Data bytes	Total length	Offset	MF
1	0 – 1479	1500	0	1
2	1480 – 2959	1500	1480/8 = 185	1
3	2960 – 3979	1040	2960/8 = 370	**0**

Example 2: the 8-byte trap ✍ MTU 1000, datagram 2400 B. 1000 − 20 = 980, not a multiple of 8 → use **976**. Data 2380 = 976 + 976 + 428; offsets **0, 122, 244**.

Example 3: read a fragment ✍ offset 100, HLEN 5, total length 100 → first data byte = 100 × 8 = **800**, data = 80 B, last byte **879**.

- MF = 1, offset = 0 → **first** fragment · MF = 0, offset ≠ 0 → **last** · MF = 0, offset = 0 → not fragmented
- One fragment lost → the **whole datagram** is discarded (reassembly timer)
- **Path MTU discovery**: send with DF = 1; a router that cannot forward drops it and returns ICMP type 3 code 4 "fragmentation needed" with the next-hop MTU
- **IPv6**: routers **never** fragment; only the source (Fragment header, 8 B, next header 44); too big → ICMPv6 "Packet Too Big"

Memory trick "**Offset × 8 = byte position**": 2^{13} offsets × 8 bytes = 2^{16}, which covers the 65 535-byte datagram.

Night-Before-the-Exam Sheet

Fact sheet

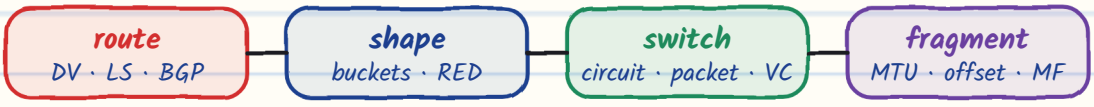
DV: $D_x(y) = \min\{c(x,v) + D_v(y)\}$
LS: flood LSAs, then Dijkstra $O(N^2)$
RIP: hops, 15 max, 16 = ∞, 30 s, UDP 520
OSPF: IP proto 89, area 0, AD 110
BGP: path vector, TCP 179, AS_PATH
AD: static 1, eBGP 20, OSPF 110, RIP 120
Hierarchy: ln N levels, e·ln N entries
Token: $S = C/(M - \rho)$ · $\max = C + \rho t$
Circuit: $s + x/b + kd$
Packet: $x/b + (k - 1)p/b + kd$
ATM 53 = 5 + 48 · VPI/VCI
T1 1.544 · E1 2.048 · STM-1 155.52
MPLS: label 20 · TC 3 · S 1 · TTL 8
Offset = byte pos / 8 (13 bits)

→ Top 14 one-liners for MCQs

1. Router = layer 3, bridge/switch = 2, hub/repeater = 1
2. Distance vector uses Bellman-Ford
3. Link state uses Dijkstra
4. BGP is the inter-AS (EGP) protocol
5. Split horizon fights count-to-infinity
6. Optimal routes to one node form a sink tree
7. Flooding tries every path: robust, many duplicates
8. Leaky bucket → constant output rate
9. Token bucket → bursts allowed, keeps tokens
10. RED drops early, at random, on avg queue
11. X.25 = hop-by-hop error control
12. Frame Relay uses DLCI; ATM uses VPI/VCI
13. IPv4 reassembly only at the destination
14. IPv6 routers never fragment

Top traps metric ≠ administrative distance · split horizon fails for 3+ router loops · switch splits collision, not broadcast, domains · virtual circuit = packet switching · leaky bucket drops packets, token bucket drops tokens · offset counts 8-byte units, and data per fragment must be a multiple of 8 · ATM HEC covers the header only

All tricks together RIP counts hops, OSPF counts cost, BGP counts ASes · OSPF "89 · 110 · area 0 · 5/6 · 10/40" · RPF = came in on my road home? · leaky = dripping tap, token = savings account · X.25 checks every hop, FR none, ATM chops into 53 · offset × 8 = byte position



the four questions every network-layer paper asks

✓ Practice MCQs (exam pattern, 10 marks)

try first, then check

1. Distance vector routing is based on:
(a) Dijkstra (b) Prim
(c) Bellman-Ford (d) Kruskal

2. In RIP, a hop count of 16 means:
(a) unreachable (b) best route
(c) 16 routers (d) default route

3. OSPF is carried directly in IP with protocol number:
(a) 6 (b) 17
(c) 520 (d) 89

4. Which protocol routes between autonomous systems?
(a) BGP (b) RIP
(c) OSPF (d) IS-IS

5. The leaky bucket algorithm mainly provides:
(a) shortest paths (b) constant output rate
(c) error detection (d) address resolution

6. Token bucket: C = 250 KB, ρ = 2 MB/s, M = 25 MB/s. Burst length ≈
(a) 125 ms (b) 10 ms
(c) 11 ms (d) 23 ms

7. Which device splits broadcast domains?
(a) hub (b) router
(c) switch (d) repeater

8. An ATM cell is:
(a) 48 bytes (b) 64 bytes
(c) 1500 bytes (d) 53 bytes

9. 4000-byte datagram (20 B header), MTU 1500: offset of the last fragment?
(a) 370 (b) 2960
(c) 185 (d) 3

10. Hop-by-hop error control is a feature of:
(a) Frame Relay (b) X.25
(c) ATM (d) MPLS

Answer key → 1-c · 2-a · 3-d · 4-a · 5-b · 6-c · 7-b · 8-d · 9-a · 10-b

Written question Explain the count-to-infinity problem of distance vector routing with an example, and describe two techniques that reduce it. 8 marks

1. DV routers know only neighbours' distances (Bellman-Ford), so they cannot tell whether a neighbour's route passes back through themselves 1
2. Chain A–B–C, costs 1. B: A = 1; C: A = 2 (via B). Link A–B fails 1
3. B hears C's "A = 2", sets A = 3 via C; C then sets A = 4 via B; they count up 5, 6, … to ∞ (16 in RIP), with a routing loop the whole time 3
4. **Split horizon**: C does not advertise A back to B, the line it learned it from; **poison reverse**: it advertises A = ∞ to B 2
5. Also hold-down timers and triggered updates; none fully solves loops of 3+ routers, which is why link state is preferred 1

Before the exam ✓ Dijkstra on a 5-node graph as a table · a DV update with Bellman-Ford · $S = C/(M - \rho)$ and $C + \rho t$ · fragment sizes, offsets, MF · also write: DV vs LS, leaky vs token bucket, X.25 vs FR vs ATM